SPARE PARTS CALCULATION

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Short Title SPARE PARTS CALCULATION

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1 SCOPE

This Standard specifies the assumptions and method of calculation used to find out the initial quantity of spares., for the long term (i.e. after the first year of operation).

2 REFERENCED STANDARDS

[1] 1AA 00210 0001 BCZZA
FAILURE RATE FACTOR

[2] 1AA 00210 0002 PHZZA
RELIABILITY CLASSES

[3] 1AA 00210 0003 BCZZA
REFERENCE FAILURE RATE

3 GENERAL CONSIDERATIONS

Spare parts strategy is an aspect that contributes to find out the system overall maintainability.The main quantities that affect spare provisioning are:

- definition of the Least Replaceable Assembly (LRA)
- LRA failure rate

- re–stocking philosophy
- LRA turn around time
- acceptable probability of stock–out

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3.1 Definition of the Least Replaceable Assembly (LRA)

The LRA is that replaceable module at which local fault diagnosis stops and direct replacement occurs.

The larger the LRA is, the faster the diagnosis occurs but, as the size of the LRA increases, so does its cost.

It is generally assumed to provide spare parts at Printed Board Assembly level.

Exception can be done for single block equipment (like outdoor radio or desk–top units) for which the equipment itself has to be provided as a spare part.

3.2 LRA failure rate

To evaluate LRA failure rate or Mean Time Between Failures (MTBF), the method described in the following Alcatel Standards shall be used:

[4] 1AA 00210 0001 BCZZA
FAILURE RATE FACTOR

[5] 1AA 00210 0002 PHZZA
RELIABILITY CLASSES

[6] 1AA 00210 XXXX XXZZA
REFERENCE FAILURE RATES

The LRA failure rate gives the expected number of failures and is the most important parameter in finding spare quantity.

3.3 Re–stocking philosophy

3.3.1 One level store

There are three main re–stocking philosophies and all the other can be seen as a combination of these three.

1st) The store shall have enough of spares for a fixed time (for example 2 years). Only at the end of this period the initial number of spares will be restored.

2nd) The store is continually replenished.
When an LRA fails in the plant, it is substituted by an identical spare and the faulty LRA is sent to repair.
After a period of time (one week up to several months) the repaired part or a new one is again available to the store.

3rd) As 2nd plus urgent procedure.

3.3.2 2 levels stores

The very general model applies:

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- The first level of store (the ”district store”) is located close to the equipment and is used for a ’group’ of systems(exchanges, transmission equipment, ...)
- When a fault occurs the failed unit is sent to the repair center.
- The failed unit is replaced by a spare unit from the district spare parts store.
- The district store gets a new spare part from the ”central store” (2 nd level of store).
- The repaired unit arrives to the central store.

3.4 LRA turn around time

This concept applies only in the **2nd and 3rd re–stocking philosophy.**

The turn around time represents the time necessary to the faulty LRA to be repaired and to return to the store again.

This time includes logistic delays (transportation) and repair necessary to deliver the LRA to the repair centre and get it back.

3.5 Acceptable probability of stock–out

This value represents the risk that the calculated number of spares is not sufficient for requests.

If the customer doesn't fix any value, this parameter can be used as a "wild card" to overestimate or underestimate the spare number.

Reasonable values for the stock–out probabilities are 1% to 5%.

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4 ONE LEVEL SPARE PARTS STORE DIMENSIONING

The methodology presented from now on applies for each type of LRA in a one level spare parts store strategy.

4.1 1st re–stocking philosophy

The store shall have a sufficient number of spares for a fixed time (see point 3.3).

PARAMETERS

n:	number of operating LRAs in the plant;
S:	number of spares necessary (to be determined) for the n LRAs;
MTBF:	LRA mean time between failures;
P_{suf}:	minimum probability that S spares are sufficient to cover the failures during the period T
T:	period of time for which the store shall have a sufficient number of spares with a probability more than $\mathbf{P_{suf}}$.
P_{out}:	Maximum shortage probability of a store of S spare units: $P_{out} = 1 - P_{suf}$

HYPOTHESIS

Predicted spares must be sufficient to cover all the time T for which reintegration is not provided.

The time to failure is exponentially distributed. This corresponds to constant failure rate, therefore failures are described by Poisson process.

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The spares are assumed not to fail while they stay in the store.

FORMULA

The necessary number of spares is determined as the minimum integer value that satisfies the formula:

$$P_{suf} \leq P = \sum_{k=0}^{S-1} \left(\frac{nT}{MTBF}\right)^{k} \frac{1}{k!} e^{-\frac{nT}{MTBF}}$$

(1') [Poisson law]

It is the probability of having less than S failures during the period T.

So, we can write:

$$P_{out} \geq 1 - \sum_{k=0}^{S-1} \left(\frac{nT}{MTBF}\right)^k \frac{1}{k!} e^{\frac{nT}{MTBF}} \quad (1)$$

4.2 2nd re–stocking philosophy

The store is continually replenished (see point 3.3).

PARAMETERS

n: number of operating LRAs in the plant

S: number of spares necessary (to be determined) for the n LRAs

MTBF: LRA mean time between failures

R: mean turn around time

P_{suf}: minimum probability that S spares are sufficient for the requests.

P_{out}: Maximum shortage probability of a store of S spare units:

$$P_{out} = 1 - P_{suf}$$

HYPOTHESIS

- Every LRA that fails in the plant is substituted by an identical one and sent to repair. In a time equal to R (on the average), the repaired part or a new one is again available to the store.
- Time to failure and restocking times are exponentially distributed. This corresponds to constant failure and restoration rates.
- Spares request at the store and reintegration of used parts happen upon statistical equilibrium condition.
- When some LRAs are lacking at the store, each of them goes back independently, the first after a mean time R.
- If the store is temporarily lacking in a requested LRA, particular ”accelerated” supplying procedures are not considered.
- The probability to ask to the store for two or more LRAs of a type in that moment lacking is negligible.
- The spares are supposed not to fail while they stay in the store.

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FORMULA

The necessary number of spares is determined as the minimum integer value that satisfies the formula:

$$P_{out} \geq \frac{\sum_{k=S}^{n+S} \alpha P_k}{\sum_{k=0}^{n+S} \alpha P_k}$$

(2)

with $\alpha = n$ for $k \leq S$
$\alpha = n+S-k$ for $k > S$

and

$$P_k = \frac{n!}{\alpha!}\, n^{\alpha - n + k}\, \frac{\left(\frac{R}{MTBF}\right)^k}{k!}\, P_0$$

$$P_0 = 1 - \sum_{k=1}^{n+S} P_k$$

It is the probability that a failure happens and that the S spare parts of the store are already used.

See APPENDIX A

4.3 3rd re–stocking philosophy

The store is continually replenished as in 4.2, but furthemore an accelerated supplying procedure is used when a shortage of store happens. This means that the store may be considered as an infinite size store.

The necessary number of spares is then determined, based on the Poisson law, as the minimum integer value that satisfies the formula

$$P_{out} \geq 1 - \sum_{k=0}^{S-1} \left(\frac{n\ R}{MTBF}\right)^k \frac{1}{k!}\ e^{-\frac{n\ R}{MTBF}}$$

(3) [Poisson law]

It is the probability that S or more than S failures happen simultaneously.

NOTES 1. – This formula (3) is the one recommended. It is commonly used due to its simplicity and because most of the operators can get urgently a spare unit from an other local store.

2. – For values of n greater or equal to 10 units and small P_{out} probabiolities ($\leq$ 5%) there is no significantdifference between formula (2) and the Poisson law (formula 3). Then formula (3) may be used even in the case of 2nd re–stocking philosophy.

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5 TWO LEVELS SPARE PARTS STORE DIMENSIONING

In this chapter we consider the spare parts dimensioning in case of two levels hierarchical spare parts store systems.

5.1 Re–stocking philosophy

The very general following model applies.

- The first level of store (the district store) is located close to the equipment and is used for a ”group” of systems (exchanges, transmission equipment, ...)
- When a fault occurs the failed unit is sent to the repair depot.
- The failed unit is replaced by a spare unit from the district spare parts.
- The district store gets a new spare part from the 2 nd level store (the central store).That takes a time L
- The repaired unit arrives at the central store after a time T_c

We assume that T_c starts at the instant when the failure happens, ie: the spare unit is immediately ”reserved”.

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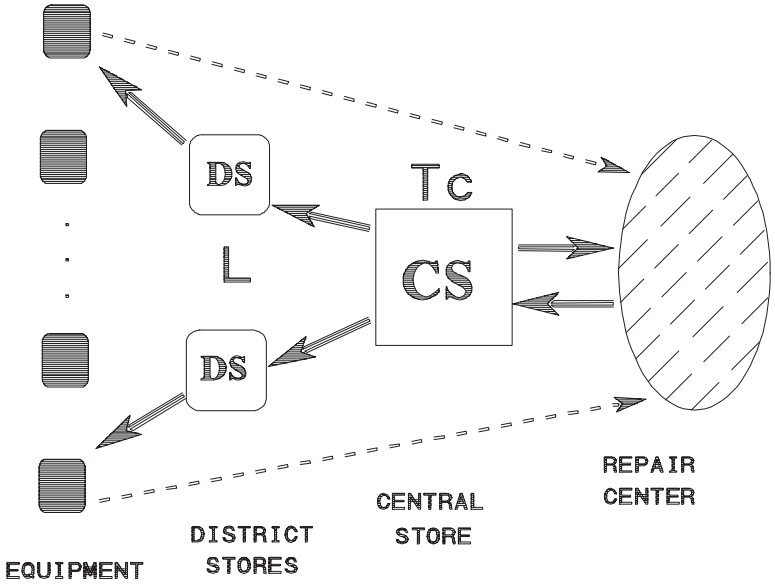


Figure 1

5.1.1 Central store dimensioning

The arrival law of the demand for spare parts, at the Central Store, may be assumed to poissonian.

For each type of LRA we have:

N: total number of units (LRA) for all the districts

MTBF: LRA mean time between failures

Sc: Number of spare parts to be found out

Tc: repair time, ie time needed to replace a spare part in the central store.

$P_{c\ out}$: maximum acceptable probability of shortage for the central store.

$P_{c\ suf}$: $P_{c\ suf} = 1 - P_{c\ out}$

The necessary number of spares is determined as the minimum integer value that satisfies the following formula:

$$P_{c\ out} \geq 1 - \sum_{k=0}^{S_c-1} \left(\frac{N\ T_c}{MTBF} \right)^k \frac{1}{k!}\ e^{-\frac{N\ T_c}{MTBF}}$$

(4) (Poisson law)

NOTES 3. – The Poisson formula may be used here because at the central store the values of S are generally large enough. Otherwise, formula (2) should be used (with T_c)

4.– In case of shortage the average delay D to get a new unit from the repair center is:

$$\mathbf{D \approx T_c / (S_c + 1)}$$

5.1.2 District store dimensioning

At the district store level, if we consider that the probability of shortage of the district store is not negligible (not very low) we must consider the reduction of the arrival rate of failed units due to no replacement (lack of spare part in the district store).

This is particularly important when the number n of LRA in each district is relatively low.

Formula (2) must be used, but with a replacement time T_D taking into account the possibility of a shortage of the central store, as defined below:

We define:

n: number of LRA (of a given type) in the district

S_D: number of spare units (of a given type) at the district store

$\lambda = 1/MTBF$: failure rate of one LRA

$\mu = 1/T_D$: replacement rate of a spare unit at the district store.

NOTE 5.– The replacement time T_D **is the time** needed to get a new spare unit from the central store. Normally,in case of no shortage,this time is L. In case of shortage, this time is L+D, with

$$D = T_c/(S_c + 1)$$

Thus, we have: $T_D = P_{c\,out}\, D + L$

If $P_{c\,out}$ is very low and S_c large enough, we can write:

$$T_D \approx L$$

The number of spare parts S_D at the district store is determined as the smallest number S that satisfies the requirement:

$$P_d \leq P_{d\,out}$$

with:

$$P_d = \frac{\sum_{k=S_D}^{n+S_D} \alpha\ P_k}{\sum_{k=0}^{n+S_D} \alpha\ P_k}$$

(5)

and with:

$\alpha = n$ for $k \leq S_D$

$\alpha = n+SD-k$ for $k > S_D$

$$P_k = \frac{n!}{\alpha!}\ n^{\alpha-n+k}\ \frac{\left(\frac{T_D}{MTBF}\right)^k}{k!}\ P_0$$

and

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$$P_0 = 1 - \sum_{k=1}^{n+S_D} P_k$$

NOTES 6.– If the number n of LRA at the district level is very large compared with the number of spare parts S_D approximatively calculated by using the Poisson law (i.e. if $n/S_D > 10$), the Poisson law may be kept to dimension the district store: that means that the influence of the lack of spare parts at the district store level is negligible on the decrease of the failure rate.

7.– Furthemore, as in the case of 1 level stock, if an urgent procedure is used to replenish the store when a shortage happens, the Poisson law must be used (formula 3)

6 GLOBAL STORE DIMENSIONING

In this chapter we consider the global store, ie. the size of the store taking into account all the types of LRA.

6.1 Mean number of shortages per year S_y

- for each type i of LRA we have:

$$s_i = P_i \, N_i \, \lambda_i \cdot 8760$$

where: s_i = shortages per year of the LRA's of type i
P_i = probability of shortage of the store of LRA of type i
N_i = number of LRA of type i
λ_i = failure rate (failures per hour) of LRA of type i

- for the global store with t types of LRAs:

$$S_y = \sum_{i=1}^{t} P_i \, N_i \, \lambda_i \cdot 8760$$

6.2 Mean time between shortages MTBS (years)

$$MTBS = \frac{1}{\sum_{i=1}^{t} P_i \, N_i \, \lambda_i}$$

6.3 Global probability of shortage P_{SG}

The probability of shortage whatever the type of LRA involved is:

$$P_{SG} = \frac{\sum_{i=1}^{t} P_i \, N_i \, \lambda_i}{\sum_{i=1}^{t} N_i \, \lambda_i}$$

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7 SPARE PARTS PROVISIONING FOR PHASE OUT COMPONENTS

If the production for a component (ASIC, IC, FET, hybrid,.....) is stopped, a special spare part supply for the installed components must be provided. A last buy of this specific component from the manufacturer must be initialized to guarantee the availability of spare parts during the useful lifetime (15 – 25 years) of the entire system.
To minimize the No. of spares and the costs, a calculation method for the number of spares parts (depending on four input parameters) is proposed in this chapter. In five steps the calculation of λ_1, λ_2 and λ_3 lead to the failure rate λ_{spare} , which is the input for the final spare parts calculation.

7.1 Input Parameters

7.1.1 Failure rate of the component

The first input parameter is the failure rate. If available, the field failure rate λ_{field} should be used to determine $\lambda_1 = \lambda_{conf}$; if not, use as basic value λpred for the calculation of λ_1.

- Confident field failure rate λ_{conf}

The failure rate is evaluated from all existing field performance statistical data pertaining to this specific component. The formula for the calculation of lfield is:

$$\lambda field = \frac{No.\ of\ defective\ components}{considered\ component\ hours}$$

The failure rate λ_{conf} is evaluated from the λ_{field} in accordance with the upper one–sided limit of 90% confidence level of X^2 distribution (compare with IEC 60605–4). If λ_{field} and thereby λ_{conf} is not available use λ_{pred} .

- Average predicted failure rate λ_{pred} .

To calculate λ_1, the average predicted failure rate λ_{pred} can be determined by the corresponding reliability class RC listed in the document 'Reliability Standard Reference Failure Rate' (1AA 002210 0003 BCZZA). The failure rate chosen must be in accordance with the environment code (GB, GF or GM) of the product.

- Assumed failure rate

If λ_{pred} is not available in the document 'Reliability Standard Reference Failure Rate', use the failure rate provided by the manufacturer or the best estimation for the input λ_{pred} .

7.1.2 Compatible PBA available

The availability of a compatible item, which can perform all functions of a non repairable unit has an important impact on the calculation. The second input parameter produces a security factor if only one special replaceable unit can be used to keep the system in operation.

- Fully CAE–compatible (HW and SW) item available

The introduction or existence of a compatible item must be coordinated with the systems department. All functions concerning HW and SW compatibility must be guaranteed. No additional spares must be foreseen to avoid critical situations caused by missing components for repair. The use of a new compatible items must be introduced to the customer.

- No compatible item available

If no compatible item is available, an additional number of spares (30%) is proposed. This avoids that missing spares (probably caused by increasing failure rate of the installed components e.g. wear out) could cause a cost intensive, new development, e.g. of a PBA or a converter.

7.1.3 Number of installed components

Depending on the number of installed components, a security factor of 20 % is proposed.

- Less than 2000

If less than 2000 components are installed, the number of calculated spare parts is usually lower than 100. Because the number of calculated spare parts is low, a security factor of 20 % is proposed, which should cover seldom problems with an increased failure rate (e.g. caused by a bad lot).

- More than 2000

If more than 2000 components are installed no addition is made in this step.

7.1.4 Costs of pre–stocking Components being phased out

Depending on the costs of providing spare parts, a factor of 10 % to 50% is proposed. Three different type of costs parameters can be chosen.

- Low

For the case of a spare part calculation for a (cheap) component in small series or special items (low runner), it is proposed to store at least 50% additional spare components. The risk of having to initiate a new development for only a few installed items should be minimized by this factor.

- Medium

For the case of providing spare parts for a medium priced pool (about 500 Euro) it is proposed to store 30% additional spare components.

- High

Usually, for high runner items, a confident failure rate is available. An increase of 10% in the number of spare parts should be sufficient.

7.2 Spare Part Calculation Formula

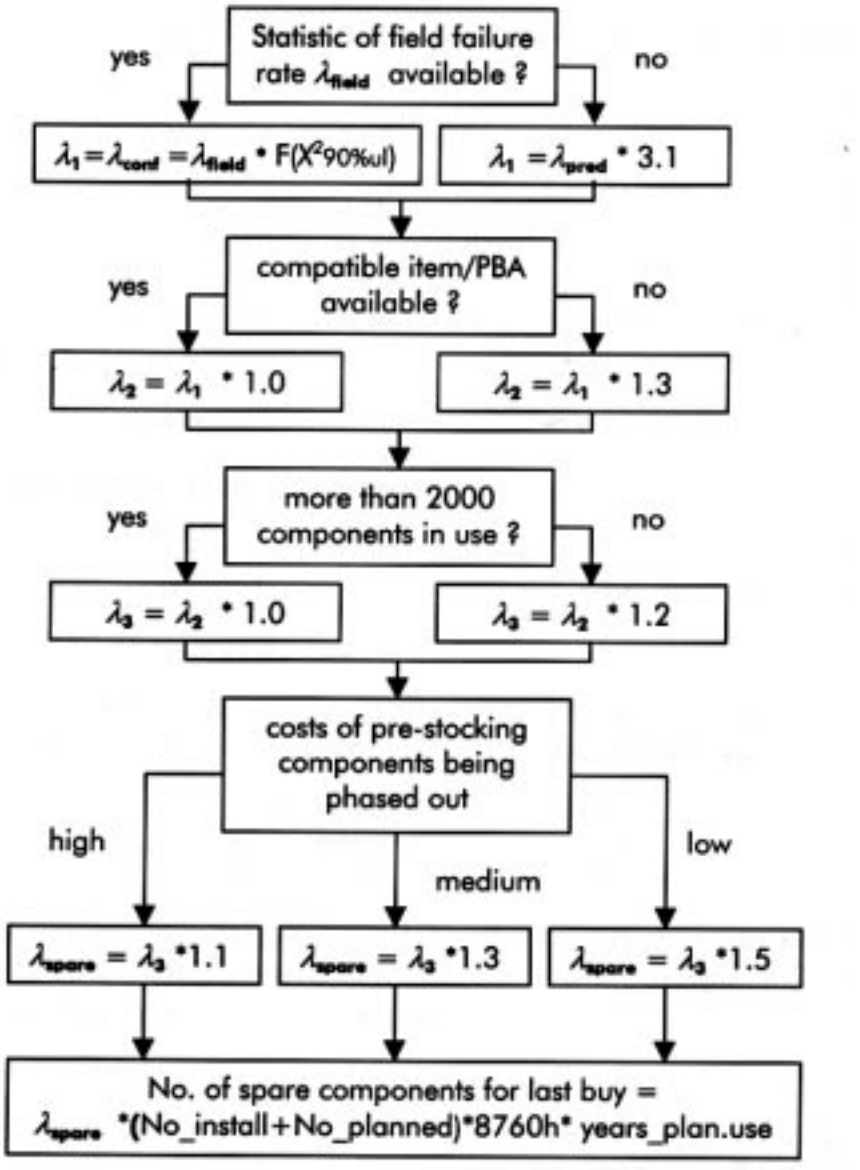
The formula for the required spare components is shown in the last step. The installed and the planned number of components to be installed in the future must be added. The planned duration of remaining useful lifetime of the system must be evaluated. The final calculation for the last buy is:

Number of spare components = λ_{spare} * total remaining item hours.

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Flow diagram for calculation
of spares for phased out components



Example:

The end of production of the of the component IC, MPCU part number 1AB 03462 0001 is announced by the manufacturer. 10600 components are installed on the PBAs DSPA. No compatible PBA is available. The price of the component is 50 Euro. In total 26 defective components were counted in the years 1998 and 1999. The PBAs should be kept in operation for a further 8 years.

λ_{field} = 26 def. components / 10600 inst. components * (8760h + 8760h) = 140 Fit

$\lambda_1 = \lambda_{conf} = \lambda_{field}$ * F ($X^2$90%ul) = 140 Fit * 1.4 = 196 Fit (factor 1.4 is interpolated from the diagram 90% confidence level; one–sided upper limit; x–axis r = 26 measured failures)

$\lambda_2 = \lambda_1$ * 1.3 = 196 Fit * 1.3 = 255 Fit (factor 1.3 => no compatible PBA available)

$\lambda_3 = \lambda_2 * 1.0 = 255$ Fit (factor 1.0 => More than 2000 components installed)

$\lambda_{spare} = \lambda_3 * 1.1 = 280$ Fit (factor 1.1 => expensive part & sufficient field data)

Number of spares = λ_{spare} * item hours = 280 Fit * 10600 inst. comp.* 8760 h * 8 = 208

The order for the last buy of the component MPCU is 208 pieces (price 10.4 kEuro).

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Appendix A

GENERAL FORMULA

We define:

n: number of LRA (of a given type) in the district

S: number of spare units (of a given type) at the store

$\lambda = 1/MTBF$: failure rate of one LRA

$\mu = 1/T_D$: replacement rate of a spare unit at the store.

NOTE The replacement time T is the time needed to get a new spare unit either from the central store or from the repair center.

The transition graph (Markow chain) is shown in figure 2:

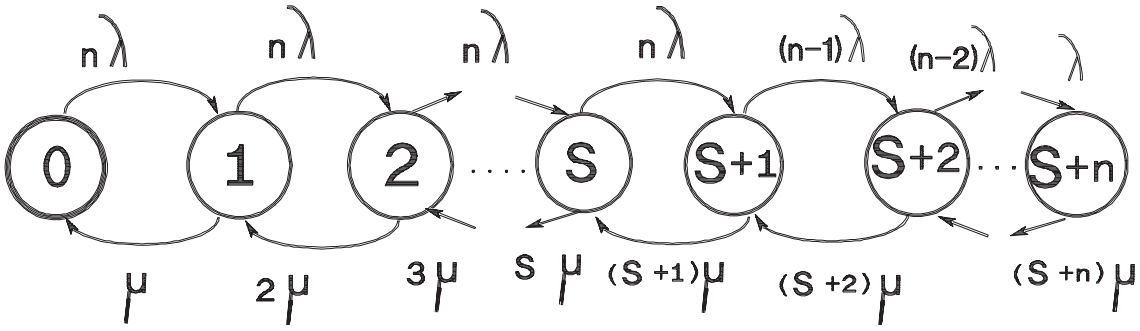


Figure 2

This system simply espresses that when the S spare units are already used, the number of possible failures becomes less than $n\lambda$ (because there is no longer replacement).

It is easily solved.

The probability of the state "k" is:

$$P_k = \frac{\left(\frac{n\lambda}{\mu}\right)^k}{k\ !} P_0 \qquad k \leq S$$

$$P_{(S+i)} = \frac{n\ !}{(n-i)\ !} \frac{S\ !}{(S+i)\ !} \left(\frac{\lambda}{\mu}\right)^i P_S$$

$$1 \leq i \leq n$$

which may be written, with k = S + **i:**

$$P_k = \frac{n\ !}{(n + S - k)\ !} \frac{S\ !}{k\ !} \left(\frac{\lambda}{\mu}\right)^{k-S} P_S$$

$$S \leq k \leq n + S$$

$$P_0 = 1 - \sum_{k=1}^{n+S} P_k$$

The expected long run fraction of demands for spares filled with delay due to shortage of spares at the district store is:

$$P_d = \frac{\sum_{k=S}^{n+S} \alpha\ P_k}{\sum_{k=0}^{n+S} \alpha\ P_k}$$

with $\alpha = n$ for $k \leq S$
$\alpha = n+S-k$ for $k > S$

The number of spare parts S_D at the district store is determined as the smallest number S that satisfies the requirement:

$$P_d \leq P_{d\,out}$$

Note From this formula it may be easily noted that when n is large, α is nearly constant; thus

$$P_d \approx \sum_{k=S}^{n+S} P_k \approx \sum_{k=S}^{\infty} P_k$$

that is the Poisson law.

- By some very simple algebric manipulations we obtain the very general expression for P_k

$$P_k = \frac{n!}{\alpha!}\ n^{\alpha - n + k}\ \frac{\left(\frac{\lambda}{\mu}\right)^k}{k!}\ P_0$$

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